PV102 Day 1 Design Lab

Applications, architecture, constraints, and technical handoff

Teams of 2	75 minutes	Design-only

Afternoon Lab | Audience: Student team | Monday, August 3, 2026

Afternoon design lab

Mission

Choose the PV architecture that best matches each client's real job. Show the energy path, identify the dominant constraint, state the safety boundary, and deliver a two-minute handoff.

Team structure

- Work in three teams of two. Start with the assigned scenario, then rotate if time permits.
- Roles: designer and verifier/recorder. Switch roles after each scenario.
- Use component cards or de-energized components only. No electrical connections are permitted.

Required deliverables

1. Architecture choice: grid-tied, hybrid, off-grid, or PV-direct.
2. A labeled energy-path diagram from source to load.
3. The critical load or operating need.
4. Three site or client constraints and the dominant constraint.
5. Protection, isolation, or labeling boundaries that must be designed before construction.
6. At least two unknowns that require verification.
7. A two-minute handoff: scope, decision, evidence, safety boundary, and unresolved items.

Acceptance criteria

Match the need. Show the path. Name the constraint. State the safety boundary. Record the unknowns.

Scenario 1 - Remote trail counter

- A 12 V electronic trail counter uses 8 W continuously.
- No utility service is available.
- The site has seasonal shade and can go five days between service visits.
- The counter must collect data day and night.

Design brief

1. Architecture choice and one-sentence reason

2. Critical load or operating need

3. Energy path - write components in order and draw arrows

4. Dominant constraint and supporting evidence

5. Two additional constraints

6. Required protection, isolation, and labeling boundaries

7. Unknowns to verify before sizing or building

8. Two-minute handoff notes

Scenario 2 - Small shop

- Utility service is available.
- The owner wants lower electric bills.
- During an outage, internet equipment and office lighting totaling about 400 W should operate for four hours.
- Roof area is available but one section receives afternoon shade.

Design brief

1. Architecture choice and one-sentence reason

2. Critical load or operating need

3. Energy path - write components in order and draw arrows

4. Dominant constraint and supporting evidence

5. Two additional constraints

6. Required protection, isolation, and labeling boundaries

7. Unknowns to verify before sizing or building

8. Two-minute handoff notes

Scenario 3 - Daytime water pump

- A 300 W DC pump fills an elevated tank.
- No utility service is available.
- The pump only needs to run when strong sunlight is available.
- The tank provides water after sunset; battery storage is not requested.

Design brief

1. Architecture choice and one-sentence reason

2. Critical load or operating need

3. Energy path - write components in order and draw arrows

4. Dominant constraint and supporting evidence

5. Two additional constraints

6. Required protection, isolation, and labeling boundaries

7. Unknowns to verify before sizing or building

8. Two-minute handoff notes

Peer verification and exit ticket

Peer verification checklist

☐ The architecture matches the client's stated operating need.

☐ The energy path includes every major required function.

☐ The dominant constraint is supported by scenario evidence.

☐ Protection and isolation are treated as design requirements.

☐ Unknowns are recorded instead of guessed.

☐ The handoff separates verified facts from open questions.

Exit ticket

1. Which operating need most strongly determines architecture?

2. When is storage unnecessary?

3. What is one question you must verify before Day 2 site assessment?